SYED DARAIN HYDER KAZMI

MACHINE LEARNING ENGINEER

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SUMMARY

AI/ML Engineer with industry experience designing, training, and deploying production machine learning systems, from computer vision models served through FastAPI to scalable data pipelines for multi-client analytics. Skilled across the full ML lifecycle in Python, PyTorch, and Scikit-Learn, from data engineering to model optimization and deployment. Applied expertise in computer vision, NLP, LLMs, and multi-agent AI.

PROFESSIONAL EXPERIENCE

AI/ML Engineer – Part-time (Syvyo) | Sep 2026 – Present

- Design, train, and deploy production computer vision and predictive ML models in PyTorch, serving them through FastAPI REST APIs as containerized microservices for real-time inference.
- Own the ML lifecycle end to end: data preprocessing, feature engineering, tuning, and evaluation.
- **Previously ML/AI Intern (Jun – Sep 2025):** built CV models and inference APIs; advanced to ML Trainee.

AI Engineer Intern (Data Pilot) | April 2026 – June 2026

- Built and maintained scalable AI/ML data pipelines for Datatram, a multi-client social media dashboard, ensuring data integrity and real-time processing via ETL/ELT workflows while collaborating cross-functionally to deliver production-ready ML components across data quality assurance, schema validation, and pipeline orchestration for multiple client environments.

PROJECTS

Adaptive Multi-Agent E-Learning System

- Built a LangGraph multi-agent tutor (5 agents) with a custom transformer knowledge-tracing model (0.693 AUC, beating DKT and BKT) served via ONNX; a Thompson-sampling bandit adapts teaching strategy and quiz difficulty in real time.

BioMed Research Helper

- Built BioAtlas, a hybrid semantic search engine over 14,693 PubMed studies: fine-tuned bge-small + BM25 fusion reaching 96.7% Recall@1, served as int8 ONNX on CPU via FastAPI, with cited evidence briefs and a React topic map.

Image Quality Assessment

- Built Lumina, a no-reference IQA system: a CLIP ConvNeXt-B with multi-scale quality, attention, and defect heads reaching 0.940 SRCC on KonIQ-10k (0.830 zero-shot on LIVE-C), served as int8 ONNX via FastAPI with a React app.

SKILLS

ML/DL Frameworks: PyTorch, Scikit-Learn, Hugging Face Transformers, NLTK, TensorFlow (familiar)
Data Engineering: Pandas, NumPy, Feature Engineering, Schema Validation
Model Development: Computer Vision, NLP, Supervised & Unsupervised Learning, Hyperparameter Tuning, Cross-Validation, Model Optimization
MLOps & Deployment: FastAPI, Streamlit, REST APIs
Tools & Platforms: Git, GitHub, Jupyter, Linux, Figma, Matplotlib, Seaborn

EDUCATION

Bachelor of Computer and Information Sciences | 2023–2027

Pakistan Institute of Engineering and Applied Sciences, Islamabad

- Relevant Courses: AI, Intro to CV, NLP, .NET, Database Management System, OOP, DSA, Networking, Software Engineering, Algorithm Analysis, Probability & Stats, Internet Application Development.

ADDITIONAL INFORMATION

- **Languages:** English (fluent), Urdu (native)
- **Co-curricular Activities:** PIEAS AI Society (President), GDGoC PIEAS (Outreach Co-Lead), PIEAS Blood Society (General Secretary), AWS Cloud Club (Graphic Designer)